Helper Functions API

ASTR 3070 Fall 2026

fetch_filenames

`fetch_filenames(dir)`

Fetches a list of filenames from a directory.

Args:
dir (str): path of the file directory

Returns:
list[str]: list of filenames

stack_master

`stack_master(files)`

Stacks frames to create a master file.
This does NOT align the frames when stacking.

Args:
files (list): list of filenames

Returns:
(np.array): final image stack in the form of a 2D array

calibrate_lights

`calibrate_lights(light, dark, flat_bias)`

Applies dark and bias/flat corrections to individual light frames.

Args:
light (list[str]): list of light frames
dark (np.array): master dark frame
flat_bias (np.array): master flat/bias frame

Returns:
list[np.array]: list of corrected light frames

align_image

`align_image(list_of_images1, list_of_images2, reference_image)`

Aligns images from multiple nights.

Args:
list_of_image_hdus (list[np.array]): list of calibrated lights from each night
reference_image (np.array): image to use as reference for alignment

Returns:
(np.array): aligned master stack array
count (int): returns the number of frames stacked